- **Internship Name : Data Analysis**
- **Task Name: Task 13: Exploring Relationship student Marks Data using Seborn**
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- **Internship Organization :**
- **Date:**

1. Objective

To create heatmaps, boxplots and distribution plots using Seaborn and use them to explore the distribution of marks and the relationships between subjects in a student marksheet.

2. Dataset Description

The dataset is an Excel file (marksheet 1.xlsx) containing student marks. Columns used for analysis:

- Subjects: Java, SEPM, ADBMS, CHN, NS (numeric marks)
- Other columns in the file: Sr.no., Student Name, Total, Average, Result, Grade
- Number of students: 16

Tools used: Python 3.13, Pandas, Seaborn, Matplotlib, Jupyter Notebook (VS Code).

3. Implementation

Step 1: Import libraries, load and prepare the data

The libraries are imported, the Excel file is loaded, the header row is detected, empty rows/columns are removed and the subject marks are converted to numbers.

```
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt

sns.set_theme(style="whitegrid")

# Excel load kara (header automatic sodhto)
df = pd.read_excel(r"C:\Users\khond\OneDrive\Desktop\Shubhu Excel sheets\marksheet 1.xlsx", header=None)

for i in range(len(df)):
    if "Java" in df.iloc[i].astype(str).str.strip().tolist():
        header_row = i
        break

df.columns = df.iloc[header_row].astype(str).str.strip()
df = df.iloc[header_row + 1:].reset_index(drop=True)
df = df.dropna(how="all")
df = df.dropna(axis=1, how="all")

# Marks numbers madhe convert kara
subjects = ["Java", "SEPM", "ADBMS", "CHN", "NS"]
df[subjects] = df[subjects].apply(pd.to_numeric, errors="coerce")

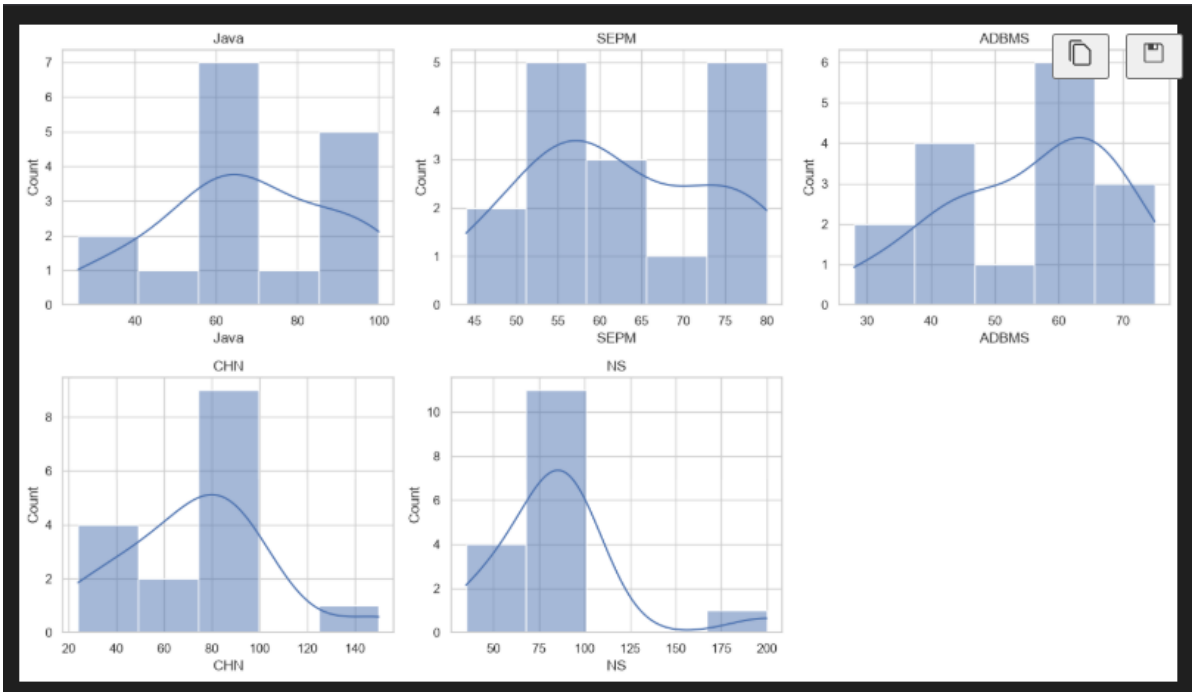
print(df[subjects].describe())
```

```
5            Java       SEPM      ADBMS         CHN          NS
count   16.000000  16.000000  16.000000   16.000000   16.000000
mean    68.375000  62.437500  55.562500   73.750000   85.875000
std     21.908522  11.575657  13.266342   31.186536   35.786171
min     26.000000  44.000000  28.000000   24.000000   35.000000
25%     58.750000  55.500000  45.000000   53.250000   74.000000
50%     65.000000  60.500000  60.000000   77.500000   87.500000
75%     87.250000  74.000000  65.000000   89.500000   91.500000
max    100.000000  80.000000  75.000000  150.000000  200.000000
```

Step 2: Distribution Plot

A histogram with a KDE curve is drawn for each subject to see how the marks are spread.

```
[9]
fig, axes = plt.subplots(2, 3, figsize=(14, 8))
for ax, sub in zip(axes.flatten(), subjects):
    sns.histplot(df[sub], kde=True, bins=5, ax=ax)
    ax.set_title(sub)
axes.flatten()[-1].axis("off")
plt.tight_layout()
plt.savefig("distribution.png", dpi=300, bbox_inches="tight")
plt.show()
```



Observations:

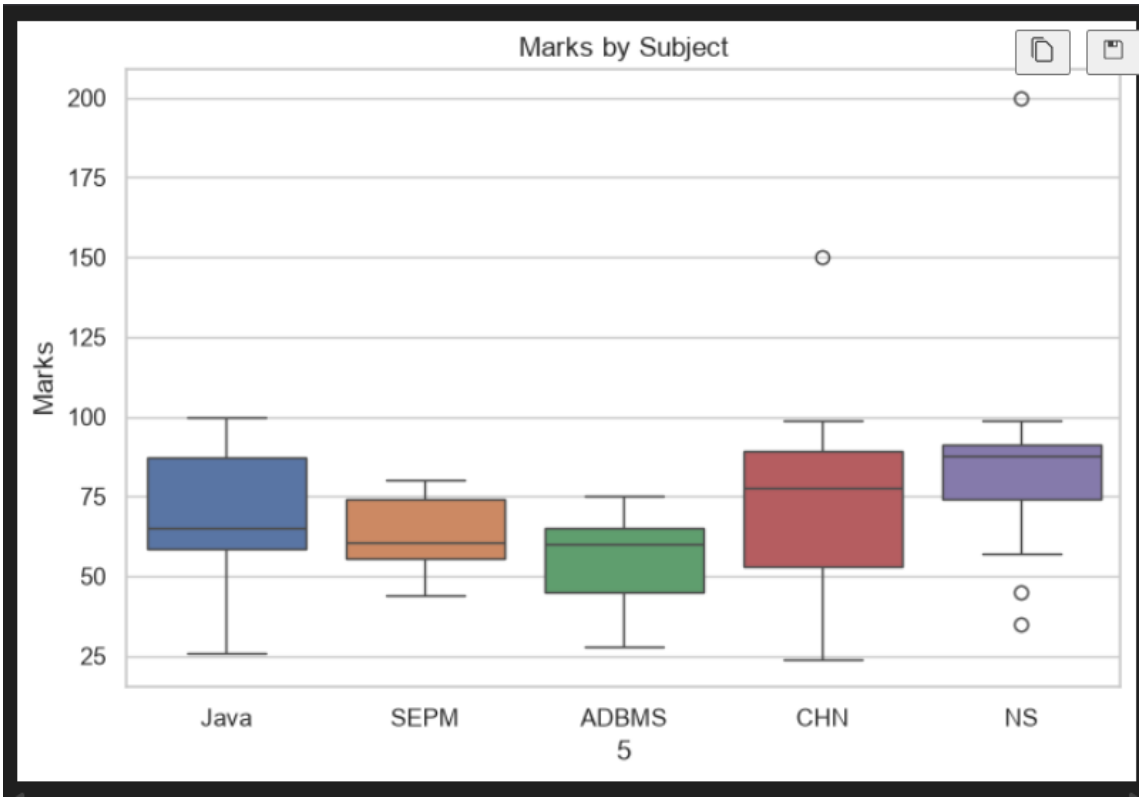
- Java: most students (7) scored between 60 and 70 marks.
- SEPM: the marks form two groups, one around 55-60 and another around 75-80.
- ADBMS: most students scored between 60 and 65 marks.
- CHN and NS: most marks are high, but a few very large values (150 and 200) stretch the plot.

Step 3: Boxplot

A boxplot is drawn for all subjects to compare their medians, spread and outliers.

```python
plt.figure(figsize=(8, 5))
sns.boxplot(data=df[subjects])
plt.title("Marks by Subject")
plt.ylabel("Marks")
plt.savefig("boxplot.png", dpi=300, bbox_inches="tight")
plt.show()
```

Python



Observations:

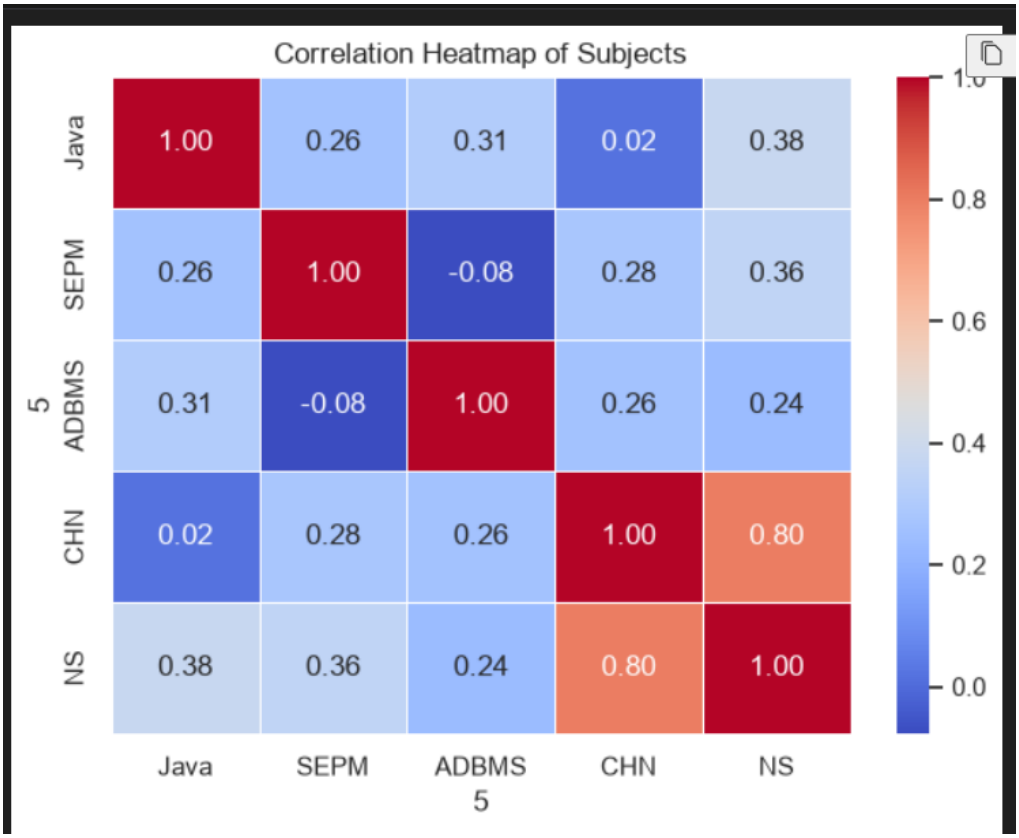
- NS has the highest median (about 87) and ADBMS has the lowest (about 60).
- SEPM has the smallest spread, so its marks are close to each other.
- Java and CHN have a wide spread, so student performance varies a lot.
- Outliers: CHN has a value of 150 and NS has a value of 200 (and two low values around 35-45). Marks above 100 are likely data entry mistakes.

Step 4: Heatmap

A correlation heatmap shows how strongly the marks of one subject are related to the marks of another. Values near +1 mean a strong positive relationship, near 0 mean no relationship and near -1 mean a negative relationship.

```
corr = df[subjects].corr()

plt.figure(figsize=(7, 5))
sns.heatmap(corr, annot=True, cmap="coolwarm", fmt=".2f", linewidths=0.5)
plt.title("Correlation Heatmap of Subjects")
plt.savefig("heatmap.png", dpi=300, bbox_inches="tight")
plt.show()
```



4. Conclusion

The distribution plots showed how marks are spread in each subject, the boxplot compared the subjects and highlighted outliers, and the heatmap showed which subjects are related. NS had the best performance and ADBMS the weakest, while SEPM marks were the most consistent. Values above 100 in CHN and NS should be checked and corrected in the source data. Seaborn made it easy to explore the data visually and find patterns that are hard to see in a plain table.